

Solution

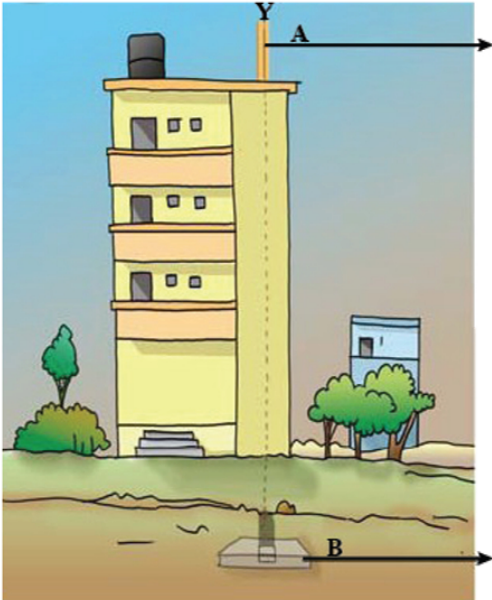
SNP-02

Class 08 - Science

Section A

1. Read the text carefully and answer the questions:

Anuj is studying about safety measures that one should take during lightning and thunderstorm. Hearing thunder is an alert to rush to a safer place. After hearing the last thunder, wait for some time before coming out of the safe place. He also read what should we do and don't. And he also study about the Lightning conductor.



- (i) **(b)** Lightning conductor

Explanation: Lightning conductor

- (ii) **(a)** To protect the buildings

Explanation: To protect the buildings

- (iii) **(b)** All of these

Explanation: All of these

- (iv) 1. Copper Plate

- (v) **(a)** True

Explanation: True

2. Read the text carefully and answer the questions:

Two group of students carrying the following activity

Step 1 - Inflate two balloons. Hang them in such a way that they do not touch each other.

Step 2 - Rub both the balloons with a woollen cloth and release them.



- (i) **(b)** Repel each other

Explanation: Repel each other

- (ii) **(d)** 2

Explanation: 2

- (iii) **(a)** Attract each other

Explanation: Attract each other

- (iv) 1. motion

- (v) (a) True

Explanation: True

3. Read the text carefully and answer the questions:

Natural phenomena can cause large scale destruction of human life and property. Fortunately, these phenomena can be predicted to some extent. The weather department can warn about a thunderstorm developing in some area. Major earthquakes are much less frequent. They can cause immense damage to buildings, bridges, dams and people. There can be a great loss to life and property.

- (i) (a) Both Uri and Tangdhar

Explanation: Both Uri and Tangdhar

- (ii) (b) Eurasian plate

Explanation: Eurasian plate

- (iii) (a) 26 January 2001

Explanation: 26 January 2001

- (iv) 1. Richter scale

- (v) (a) True

Explanation: True

4. Read the text carefully and answer the questions:

Step 1 - Take an empty jam bottle. Take a piece of cardboard slightly bigger in size than the mouth of the bottle.

Step 2 - Pierce a hole in it so that a metal paper clip can be inserted.

Step 3 - Cut two strips of aluminium foil about $4\text{ cm} \times 1\text{ cm}$ each. Hang them on the paper clip as shown.

Step 4 - Insert the paper clip in the cardboard lid so that it is perpendicular to it.

Step 5 - Charge a refill and touch it with the end of the paper clip.



- (i) (d) Earthing

Explanation: Earthing

- (ii) (d) Protect from electrical shocks due to any leakage of electrical current.

Explanation: Protect from electrical shocks due to any leakage of electrical current.

- (iii) (a) Coulomb

Explanation: Coulomb

- (iv) 1. Electroscope

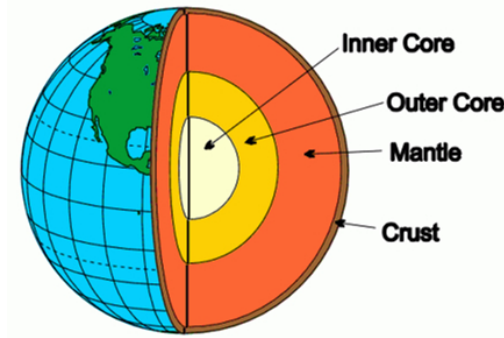
- (v) (a) True

Explanation: True

5. Read the text carefully and answer the questions:

In ancient times, people did not know the true cause of earthquakes. The tremors are caused by the disturbance deep down inside the uppermost layer of the earth called the crust. The outermost layer of the earth is not in one piece. Tremors on the earth can also

be caused when a volcano erupts, or a meteor hits the earth, or an underground nuclear explosion is carried out.



- (i) **(c)** Earthquake

Explanation: Earthquake

- (ii) **(a)** 8.0

Explanation: 8.0

- (iii) **(b)** Richter scale

Explanation: Richter scale

- (iv) 1. Fragmented

- (v) **(b)** False

Explanation: False

Section B

6.

- (c)** A copper rod

Explanation: Only non-conducting materials can be easily charged by friction. Copper is a highly conducting material. Therefore, a copper rod cannot be charged easily by friction.

7.

- (c)** Air inside expands

Explanation: A candle flame produces heat. When a balloon is taken near candle, the heat causes the air molecules inside the balloon expands that leads to bursting of balloon due to high pressure inside it. The heated air exerts pressure on the walls of the balloon and due to uneven expansion the balloon bursts.

8.

- (c)** Tornado

Explanation: A tornado is a violent storm with a column of rapidly rotating winds, having the appearance of a dark, funnel-shaped cloud that reaches from the sky to the ground. It causes much damage from the sheer force of its high speed winds.

9.

- (d)** Electron

Explanation: An object becomes charged when the atoms in the object gain or lose electron. The object that gains electron becomes negatively charged and one who loose electron becomes positively charged.

10.

- (d)** Induction

Explanation: Rearranging charges in uncharged metal object without contact with charged object is called induction because it does not required any contact but charge jumps from high end to the lower end.

11.

- (d)** On the top of building

Explanation: On the top of building

12.

- (a)** They occurs suddenly

Explanation: Natural disaster cannot be predicted in advance because natural disaster occurs suddenly although a warning or predication of its occurrence is nowadays is given with the help of the pictures sent by weather satellites.

13.

- (c)** Lightening rod at the top of building

Explanation: Building can be protected from lightening by using lightening rod/ conductor at the top of building. Lightening conductor is placed at highest point of the building which can easily attract the free ions passing nearby and transfer it to earth.

14.

(c) Light travels faster than sound

Explanation: Light travels at the speed of 3×10^8 m/s while sound travels at a speed of 332 m/s i.e., light travels much faster than sound that is why lightning is seen earlier and thunder is heard later.

Section C

15.

(d) A is false but R is true.

Explanation: Trees can get hit by lightning very easily. Also trees that are alone have more probability of getting hit by lightning. Therefore, it is not safe to stand under a tree during lightning.

So, (A) is the false, and (R) is the correct statement.

16.

(d) A is false but R is true.

Explanation: The whole crust of the earth is not in one piece. It is divided into many plates.

The power of an earthquake is expressed in terms of a magnitude on a scale called Richter scale.

So, (A) is the false and (R) is the correct statement.

17.

(a) Both A and R are true and R is the correct explanation of A.

Explanation: An earthquake is a sudden shaking or trembling of the earth lasting for a very short period of time. It is caused by a disturbance deep inside the earth's crust.

So, (A) and (R) are the true statement and (R) is the reason of (A).

18.

(c) A is true but R is false.

Explanation: Electrical charge can be transferred from one object to another object through conductors. because conductors have free electrons that carry charges from one object to another object. Metal is a good conductor of electricity.

So, (A) is the true and (R) is the false statement.

19.

(b) Both A and R are true but R is not the correct explanation of A.

Explanation: Static electricity can be created by rubbing one object against another object.

A flow of electric charge constitutes an electric current. So, when charges move they constitute an electric current.

So, (A) and (R) are the true statement and (R) is not the reason for (A).

20.

(b) Both A and R are true but R is not the correct explanation of A.

Explanation: Lightning is caused due to the accumulation of electric charge in the clouds.

In highly seismic areas, the use of mud or timber is better than heavy construction material. We should keep the roofs as light as possible.

So, (A) and (R) are the true statement and (R) is not the reason for (A).

21.

(b) Both A and R are true but R is not the correct explanation of A.

Explanation: Lightning Safety Lightning strikes could destroy life and property. It is, therefore, necessary to take measures to protect ourselves.

A lightning conductor does not allow the charge to accumulate on a building. It transfers all the charges to the earth, protecting the building from being struck by lightning.

So, (A) and (R) are the true statement, and (R) is not the reason for (A).

22.

(c) A is true but R is false.

Explanation: A lightning Conductor is a device used to protect buildings from the effect of lightning. Whenever lightning falls on the building, the electric charges are carried by the conductor into the ground and hence, the building remains unaffected.

And direct lightning strike is not useful.

So, (A) is the true and (R) is the false statement.

23. **(c)** A is true but R is false.
Explanation: The innermost part of the earth is called it's core. The outermost part of the earth is called the crust. The central region of the earth is called the mantle.
 So, (A) is the true and (R) is the false statement.
24. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Volcanic eruptions, explosions, landslides, avalanches, and even rushing rivers can also cause seismic waves. Seismic waves travel through and around the Earth and can be recorded with seismometers.
 So, (A) and (R) are the true statement and (R) is the reason for (A).

Section D

25. State True or False:

- (i) **(a)** True
Explanation: True
- (ii) **(a)** True
Explanation: True
- (iii) **(a)** True
Explanation: True
- (iv) **(b)** False
Explanation: False
- (v) **(a)** True
Explanation: True
- (vi) **(b)** False
Explanation: False
- (vii) **(b)** False
Explanation: False
- (viii) **(a)** True
Explanation: True
- (ix) **(b)** False
Explanation: False
- (x) **(b)** False
Explanation: False
- (xi) **(a)** True
Explanation: True
- (xii) **(b)** False
Explanation: False
- (xiii) **(b)** False
Explanation: False
- (xiv) **(a)** True
Explanation: True
- (xv) **(b)** False
Explanation: False
- (xvi) **(b)** True
Explanation: True
- (xvii) **(a)** True
Explanation: True
- (xviii) **(a)** True
Explanation: True

(xix) (a) True

Explanation: True

(xx) (a) True

Explanation: True

Section E

26.	Column A	Column B
	(i) Richter Scale	(d) earthquake
	(ii) Earthquake	(e) shaking of earth
	(iii) Resin	(a) amber
	(iv) Thunderstorm	(b) meeting of (+ve) and (-ve) charges
	(v) Lightning rod	(c) Protect houses
	(vi) Electroscope	(f) charge detecting device

27. i. - b

ii. - a

iii. - d

iv. - c

28. i. - c

ii. - b

iii. - d

iv. - a

29. i. - d

ii. - c

iii. - b

iv. - a

30. i. - b

ii. - a

iii. - d

iv. - c

31. i. - d

ii. - c

iii. - b

iv. - a

Section F

32. (a) Lightning conductor

Explanation: Skyscrapers can be protected from lightning by providing a lightning conductor at the top of the building. Lightning conductor consists of a long thick metal rod /strip having sharp spikes at its upper end. The lower end of the metal strip is connected to a large copper plate buried deep inside called the earthing, which discharges the lightning safe into the ground.

33.

(c) Electric discharge

Explanation: During lightning, electric discharge takes place between oppositely charged clouds. It is actually a great electric spark in the sky produced by the electric charges in the sky.

34.

(d) Rubbing

Explanation: Some object can be charged by rubbing with other objects. During rubbing charged object transfers its charge to another uncharged body.

35. (a) Electric spark

Explanation: Lightning occurs when two oppositely charged ions present in clouds come in contact with each other that

cause electric spark.

36.

(d) Lighter

Explanation: Warm air is lighter than cold air. Warm air moves upward leaving low pressure below where cool air takes place.